



TCC805x MCU BSP SDK

Getting Started

Rev. 1.01 [A]

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1 INTRODUCTION

This document provides a usage of TCC805x MCU BSP SDK as follows:

- Board description (TCC805x Evaluation Board)
- Build guide for GCC and Green Hills Multi IDE
- How to make S-NOR image
- FWDN guide for downloading image
- How to debug by using JTAG debugger

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

EVb	Evaluation Board
F/W	Firmware
FWDN	Firmware Downloader PC tool by Telechips
GCC	GNU Compiler Collection
Green Hills IDE	IDE (Integrated Development Environment) by Green Hills software
BU	Battery Unit
JTAG	Joint Test Action Group
T32 (Trace32)	One of JTAG debugger
S-NOR	Serial NOR flash

3 BOARD DESCRIPTION

This chapter describes TCC805x Evaluation Board (hereinafter referred to as EVB). TCC805x MCU BSP can be executed on both TCC8059 and TCC8050_53 EVB.

3.1 EVB Description

3.1.1 CPU Board

The figure below shows TCC8059 and TCC8050_53 CPU Board. TCC805x MCU BSP was tested under CPU Board SV0.1.

- **JTAG Port:** Port for debugging via JTAG interface
- **Boot Mode Switch:** Switches to select boot mode (S-NOR and eMMC boot mode or USB boot mode)
- **FWDN MicroAB Port:** MicroAB port to connect with FWDN PC Tool
- **Reset Button:** Button for reset

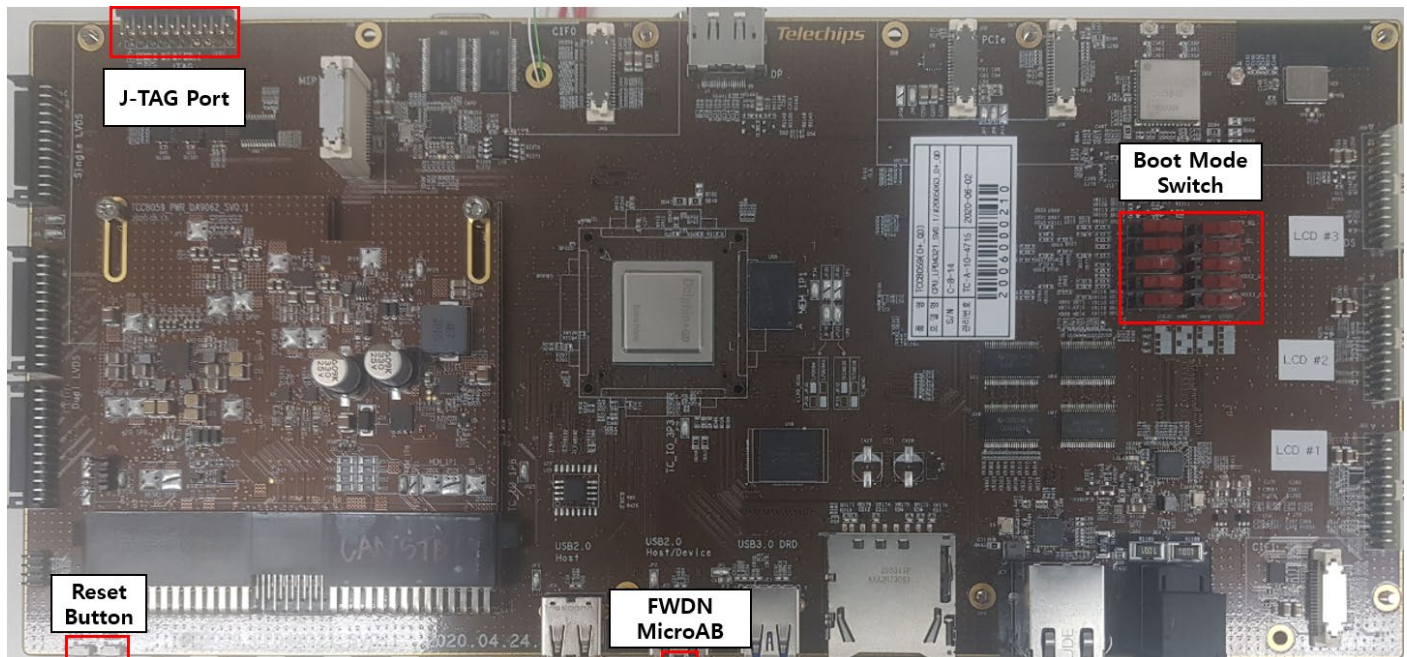


Figure 3.1 TCC8059 CPU Board

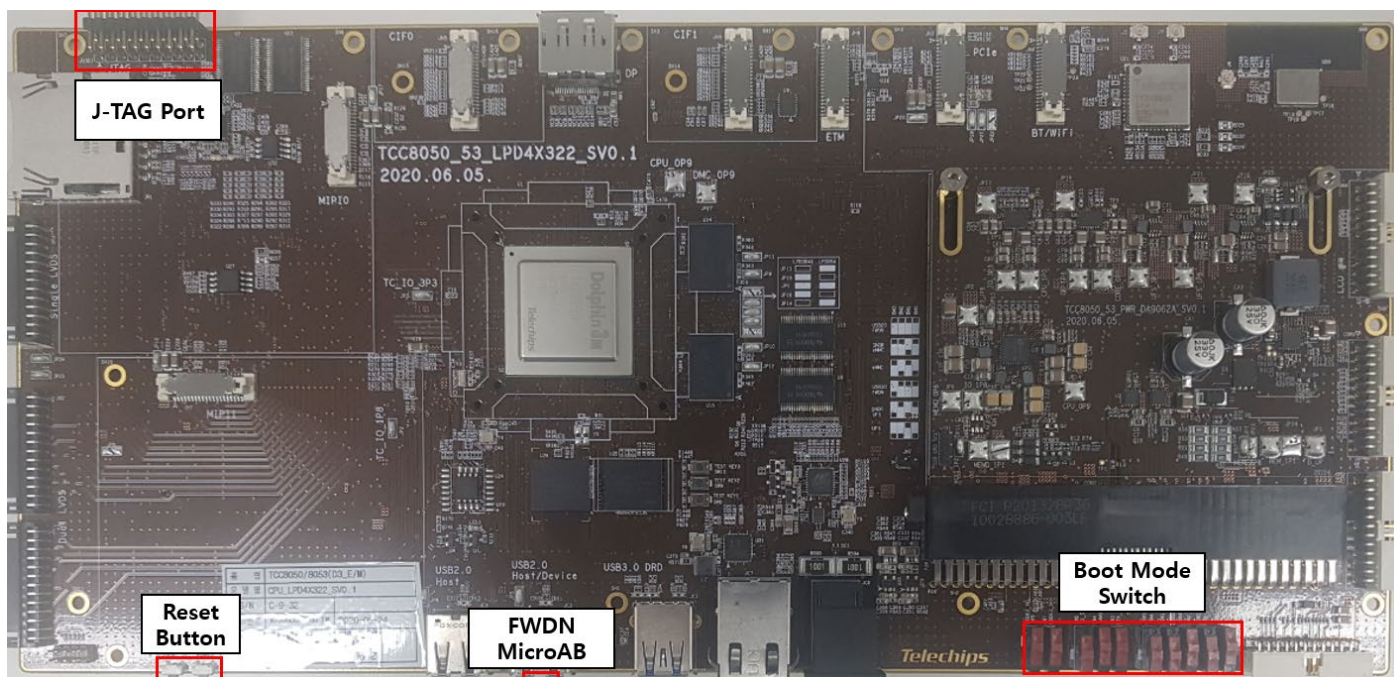


Figure 3.2 TCC8050_53 CPU Board

3.1.2 Main Board

The figure below shows TCC805x Main Board. TCC805x MCU BSP was tested under Main Board CV0.11.

- **BU 12V:** Main Power
- **UART Debug Port:** UART debug port for MCU and AP (Application Processor) UART log

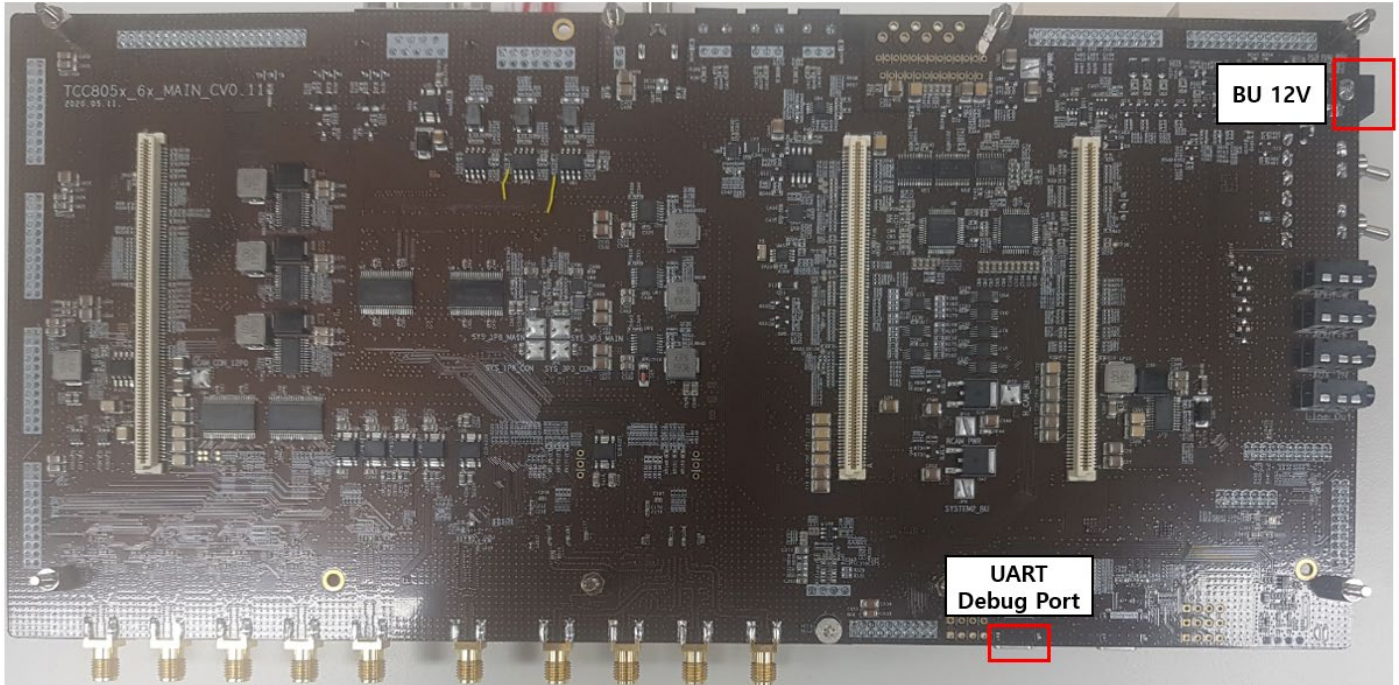


Figure 3.3 TCC805x Main Board

For more detailed information about TCC805x EVB usage, refer to "*TCC805x Common Hardware-Quick Start Guide for TCC8050_53 EVB*" or "*TCC8059 Common Hardware-Quick Start Guide for EVB*"

3.2 Boot Mode

The TCC805x EVB has boot mode switch for booting configuration. The TCC805x MCU BSP needs S-NOR and eMMC boot mode and USB boot mode for executing F/W and FWDN. For more detailed information about TCC805x EVB boot mode, refer to “*TCC805x Common Hardware-User Guide for TCC8050_53 EVB*” or “*TCC8059 Common Hardware-User Guide for EVB*”.

Note: eMMC boot mode is not used in this TCC805x MCU BSP.

3.2.1 TCC8059 EVB

Pin	Value	Figure
BM3(JSW1)	Low(0)	
BM2(JSW2)	Low(0)	
BM1(JSW3)	Low(0)	
BM0(JSW4)	Low(0)	

Figure 3.4 TCC8059 USB 2.0 Boot Mode

Pin	Value	Figure
BM3(JSW1)	Low(0)	
BM2(JSW2)	Low(0)	
BM1(JSW3)	High(1)	
BM0(JSW4)	Low(0)	

Figure 3.5 TCC8059 S-NOR and eMMC Boot Mode

3.2.2 TCC8050_53 EVB

Pin	Value	Figure
BM4(JSW3)	Low(0)	
BM3(JSW1)	Low(0)	
BM2(JSW4)	Low(0)	
BM1(JSW2)	Low(0)	
BM0(JSW5)	Low(0)	

Figure 3.6 TCC8050_53 USB 2.0 Boot Mode

Pin	Value	Figure
BM4(JSW3)	Low(0)	
BM3(JSW1)	Low(0)	
BM2(JSW4)	Low(0)	
BM1(JSW2)	High(1)	
BM0(JSW5)	Low(0)	

Figure 3.7 TCC8050_53 S-NOR and eMMC Boot Mode

4 DOWNLOAD MCU BSP

This chapter describes how to download the MCU BSP package. You can get the MCU BSP package as follows:

- Prepare git environment.
- Clone repository using “clone” command as follows: `git clone ssh://git.telechips.com/mcu/mcu_bsp.git`
- It needs access permission to clone the release git repository. If you have permission problem, contact Telechips. [2]
Check all released branch using “branch” command as follows: `git branch -a`
- Move to release branch using “checkout” command as follows: `git checkout branch name`
- For more detailed information on git, refer to git manual. You can get it by using “git help” command.

5 BUILD GUIDE

There are two build options as described below.

- GCC environment
 - This is default build environment.
 - Free license
- Green Hills IDE environment
 - Windows dedicated build environment.
 - No Free license

5.1 GCC Environment

Important: The normal operation of MCU BSP is guaranteed only in Green Hills IDE Environment.

5.1.1 Install Toolchain

Basically, MCU BSP uses "*gcc-linaro-7.2.1-2017.11-x86_64_arm-eabi*". Refer to the Makefile.

mcu_bsp/sources/build/tcc805x/gcc/Makefile

```
MCU_BSP_TOOLCHAIN_PATH      ?= /opt/gcc-linaro-7.2.1-2017.11-x86_64_arm-eabi
MCU_BSP_TOOLCHAIN_PREFIX    ?= $(MCU_BSP_TOOLCHAIN_PATH)/bin/arm-eabi-

LIBCDIR      := $(MCU_BSP_TOOLCHAIN_PATH)/arm-eabi/libc/usr/lib
LIBGCCDIR    := $(MCU_BSP_TOOLCHAIN_PATH)/lib/gcc/arm-eabi/7.2.1
LIBGCC       := -L$(LIBCDIR) -L$(LIBGCCDIR) -lc -lm -lgcc
AS           := $(MCU_BSP_TOOLCHAIN_PREFIX)as
CC           := $(MCU_BSP_TOOLCHAIN_PREFIX)gcc
LD           := $(MCU_BSP_TOOLCHAIN_PREFIX)ld
SIZE        := $(MCU_BSP_TOOLCHAIN_PREFIX)size
OBJCOPY     := $(MCU_BSP_TOOLCHAIN_PREFIX)objcopy
OBJDUMP     := $(MCU_BSP_TOOLCHAIN_PREFIX)objdump
INCLUDES    += -I$(MCU_BSP_BUILD_ROOT_PATH)
INCLUDES    += -Iinclude
INCLUDES    += -I$(LIBCDIR)/../include
INCLUDES    += -I$(LIBGCCDIR)/include
```

Figure 5.1 Toolchain for GCC Environment

5.1.2 How to build MCU BSP using GNU Make Environment

1. Go to "*mcu_bsp/sources/build/tcc805x/gcc*" path
2. Edit Makefile if you need to change build options. Refer to the Build Flags in Makefile.

mcu_bsp/sources/build/tcc805x/gcc/Makefile

```
# Build Flags
MCU_BSP_BUILD_FLAGS_VERBOSE           ?= 0
MCU_BSP_BUILD_FLAGS_DEBUG              ?= 1

MCU_BSP_BUILD_FLAGS_TARGET_OS_FREERTOS ?= 1
MCU_BSP_BUILD_FLAGS_TARGET_OS_UCOS    ?= 0
MCU_BSP_BUILD_FLAGS_TARGET_OS_NONE    ?= 0

MCU_BSP_BUILD_FLAGS_OS_CPU_VFP_NONE   ?= 1
MCU_BSP_BUILD_FLAGS_OS_CPU_VFP_D16    ?= 0
MCU_BSP_BUILD_FLAGS_SNOR_BOOT         ?= 1

MCU_BSP_BUILD_FLAGS_SOC_REVISION_ES    ?= 1
MCU_BSP_BUILD_FLAGS_SOC_REVISION_CS    ?= 0

MCU_BSP_BUILD_FLAGS_SELECT_MEMORY_SRAM_0 ?= 1
MCU_BSP_BUILD_FLAGS_SELECT_MEMORY_SRAM_1 ?= 0

MCU_BSP_BUILD_FLAGS_BOARD_NAME_TCC8050_EVM ?= 0
MCU_BSP_BUILD_FLAGS_BOARD_NAME_TCC8059_EVM ?= 1
```

Figure 5.2 Build Option for GCC Environment

3. Execute "make clean" to remove all object files.
4. Execute "make" to build.
5. You can see the MCU BSP output files as follows:

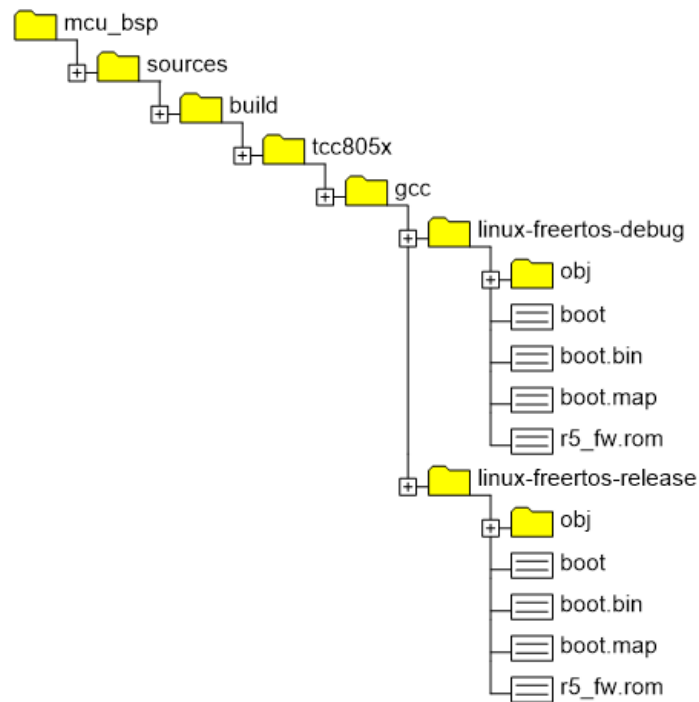


Figure 5.3 Output Files for GCC Environment

MCU BSP output files are located in "linux-freertos-release" or "linux-freertos-debug" directory. It depends on the build option (**MCU_BSP_BUILD_FLAGS_DEBUG**) in Makefile.

- A. obj: object files
- B. boot: MCU BSP elf file
- C. boot.bin: MCU BSP binary file
- D. boot.map: MCU BSP map file
- E. r5_fw.rom: Final MCU BSP rom file. It is created by adding a header to the binary file ("*boot.bin*"). It can be updated to the target EVB after being included in S-NOR image. Refer to the Chapter 5.3 Preparation Image for S-NOR Booting.

5.2 Green Hills IDE Environment

Important: MCU BSP is guaranteed only in Green Hills IDE Environment.

5.2.1 Green Hills Version

5.2.1.1 Green Hills IDE Version

The MCU BSP S/W is optimized with Green Hills MULTI Project Manager version v6.1.6 as Figure 5.4 MULTI Project Manager. The version of MULTI Project Manager can be checked via “Help→ About MULTI Project Manager” menu.

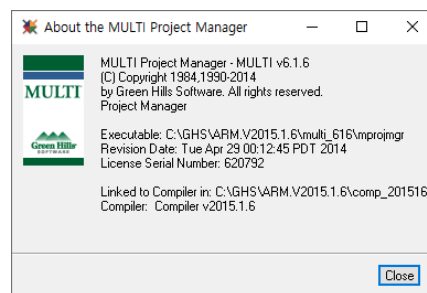


Figure 5.4 MULTI Project Manager

5.2.1.2 Green Hills Compiler Version

The following is the Green Hills compiler version of MCU BSP's. It is recommended to establish with the same version of compilers as Table 5.1 for maintaining compatibility of development environment with MCU BSP.

Table 5.1 Green Hills Compiler Version

Type	Compiler Name	Version
C Compiler	ccarm.exe	Sun Oct 18 23:01:55 PDT 2015
	ccarm64.exe	Sun Oct 18 22:53:28 PDT 2015
C++ Compiler	comp_201516/cxarm.exe	Sun Oct 18 23:00:35 PDT 2015
	comp_201516/cxarm64.exe	Sun Oct 18 22:52:18 PDT 2015
Thumb Compiler	comp_201516/ccthumb.exe	Sun Oct 18 23:01:59 PDT 2015
	comp_201516/cxthumb.exe	Sun Oct 18 23:00:39 PDT 2015
Assembler	comp_201516/asarm.exe	Sun Oct 18 22:57:43 PDT 2015
	comp_201516/asarm64.exe	Sun Oct 18 22:46:27 PDT 2015
Linker	comp_201516/elxr.exe	Sun Oct 18 22:19:08 PDT 2015
Librarian	comp_201516/ax.exe	Sun Oct 18 22:09:51 PDT 2015

5.2.2 Make New Project Using Green Hills IDE

This chapter describes the basic processor of own project setup using Green Hills IDE.

1. Write project name into "**Name**" field.

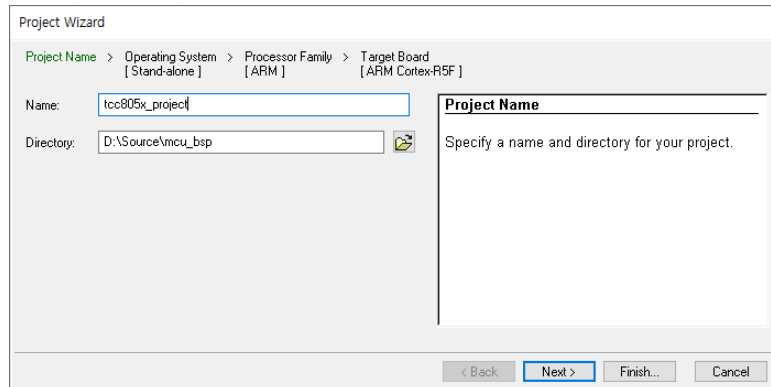


Figure 5.5 Write Project Name

2. Operating System: Select "**Stand-alone**".

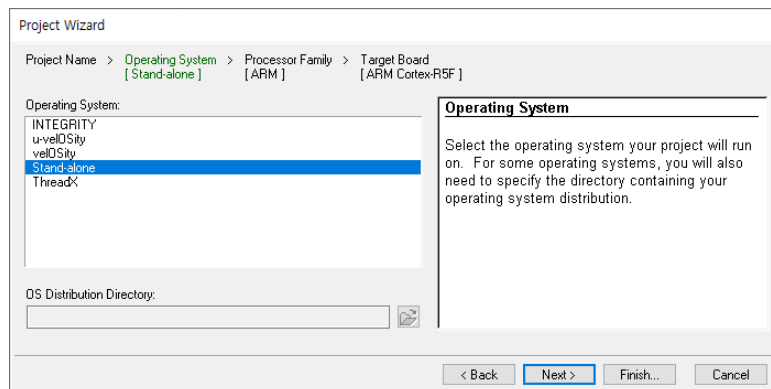


Figure 5.6 Select Operating System

3. Processor Family: Select "**ARM**".

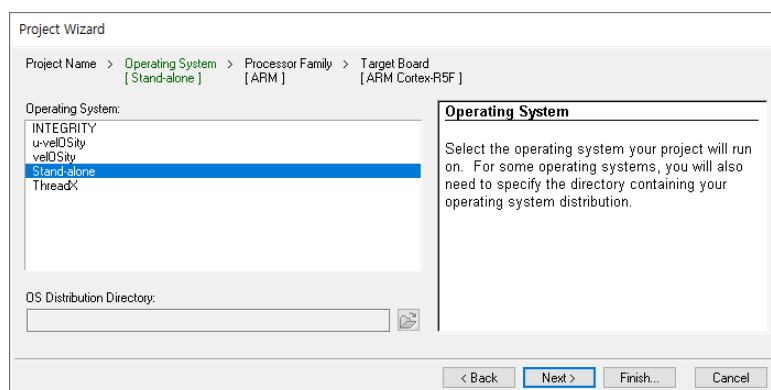


Figure 5.7 Select Processor Family

4. Target Board: Select "**Generic-ARM:ARM Cortex-R5F**".

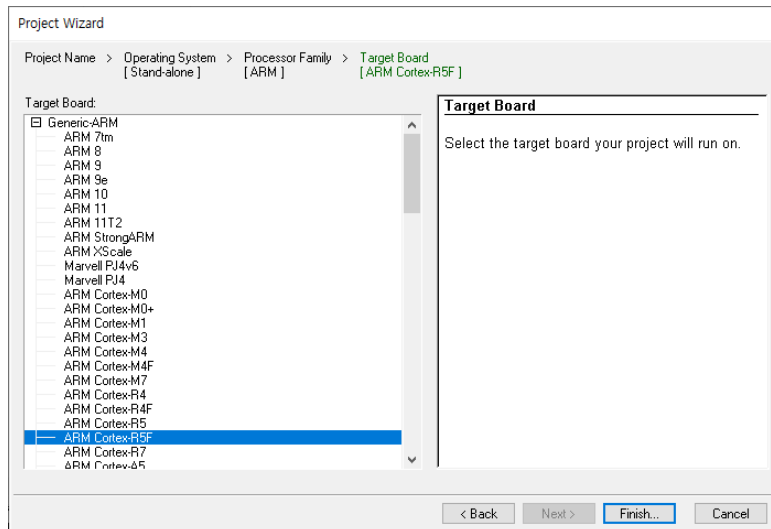


Figure 5.8 Select Target Board

5. Select "**Finish**" to activate, and then "**Select Item to Add**" Popup is shown.
 - 1) In project directory, **.gpj** files and **tgt** folder are automatically generated as shown in the Figure 5.9 Generated Project Files.

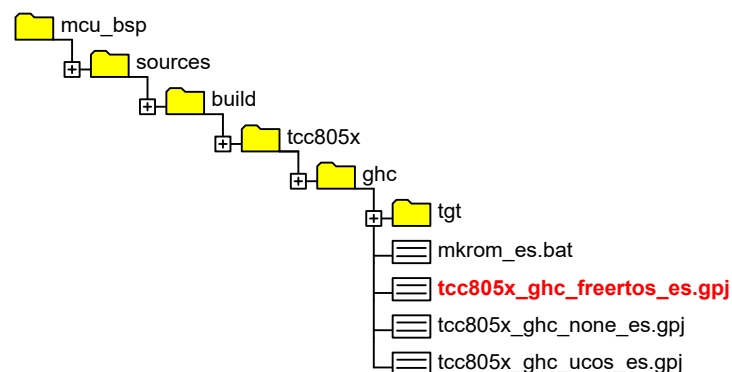


Figure 5.9 Generated Project Files

- 2) Whole setting of the project is managed in "**Project_name**".gpj file. "**Top Project**" is the highest project file.
- 3) The project file consists of top project, sub project, program, and source code files.
 - Properties of the program are as follows:
 - Individual application source + basic elements of the LD (Linker Directives) file
 - If there is a header in the project folder, it is unnecessary to add it separately (Recognizing the dependency of the header)

6. Generate a new program.

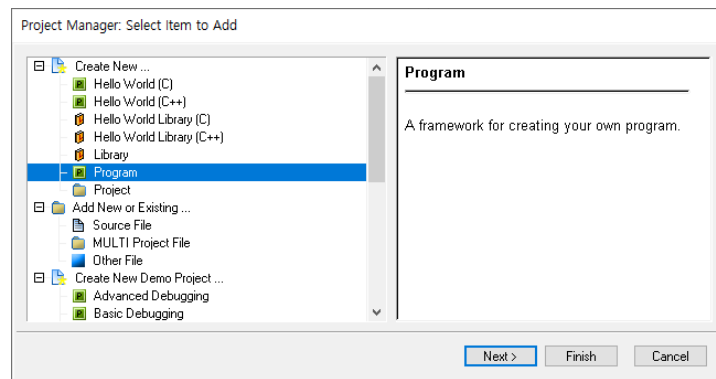


Figure 5.10 Generate New Program

- 1) Setting the source code directory and name: Select your source code directory and write project name as follows

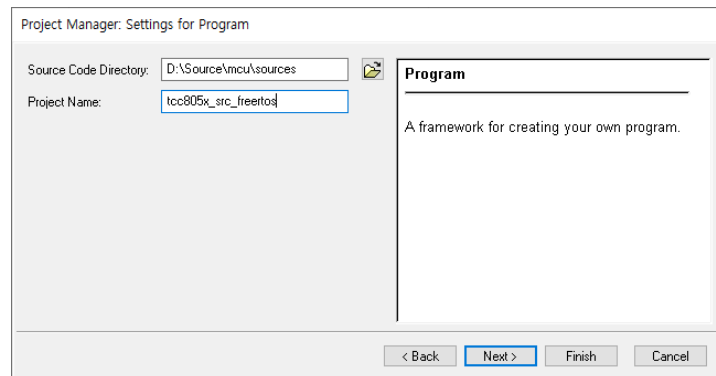


Figure 5.11 Select Source Code Directory

- 2) Setting the program layout: Set how to execute the program.

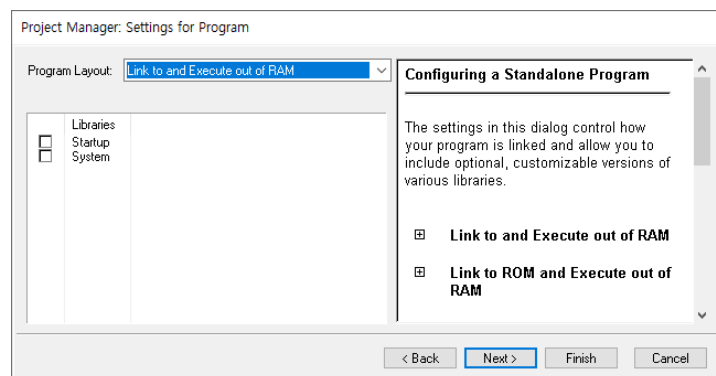


Figure 5.12 Select Program Layout

- Link to and execute out of RAM: Program code is loaded and executed in the RAM (Default setting).
- Link to ROM and execute out of RAM: Program is loaded in the ROM and executed in the RAM. It is available when the data area is copied to RAM in the startup code.

- Libraries are not selected (with own startup code).

3) Completion: Initial page is shown as follows.

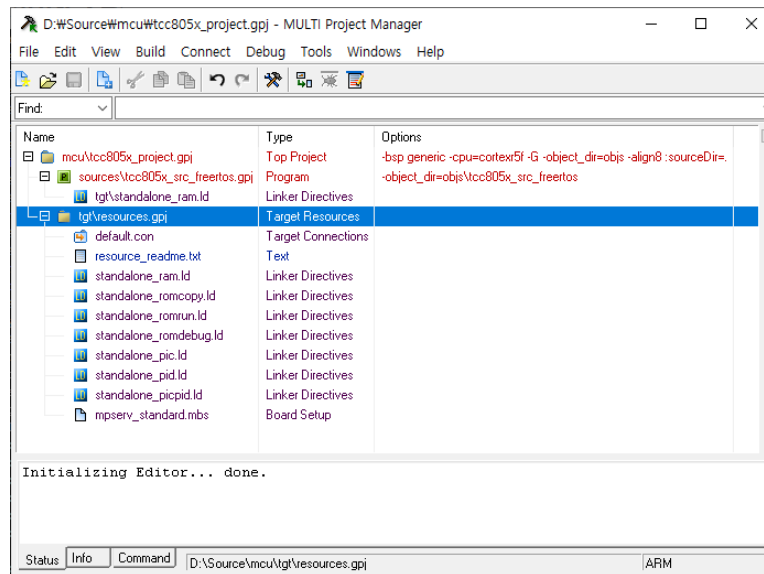


Figure 5.13 Completion of Initial Project Setup

7. Add Source Code.

- 1) Select program → Select , right-click, and select **"Add Item"** menu.

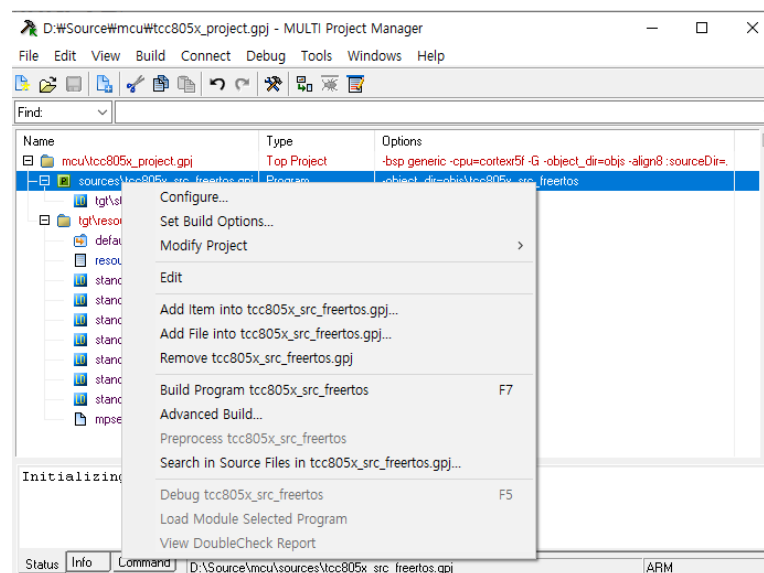


Figure 5.14 Add New Source Code Item

- 2) In **"Select Item to Add"** window → **"Add New or Existing"** → Add file in **"Source File"**.

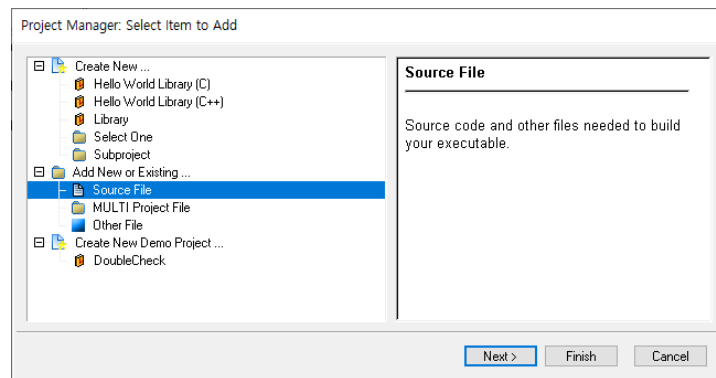


Figure 5.15 Add Source File

3) When you add the existing module individually or select the folder, ./h file is automatically added. You can also add your own modules in new project using drag and drop method.

- Search all of implementation file(.c) using wild character(*.c) on Windows browser.
- Select the files and drag and drop under program file (tcc805x_src_freertos.gpj).
- The result after adding is as follows.

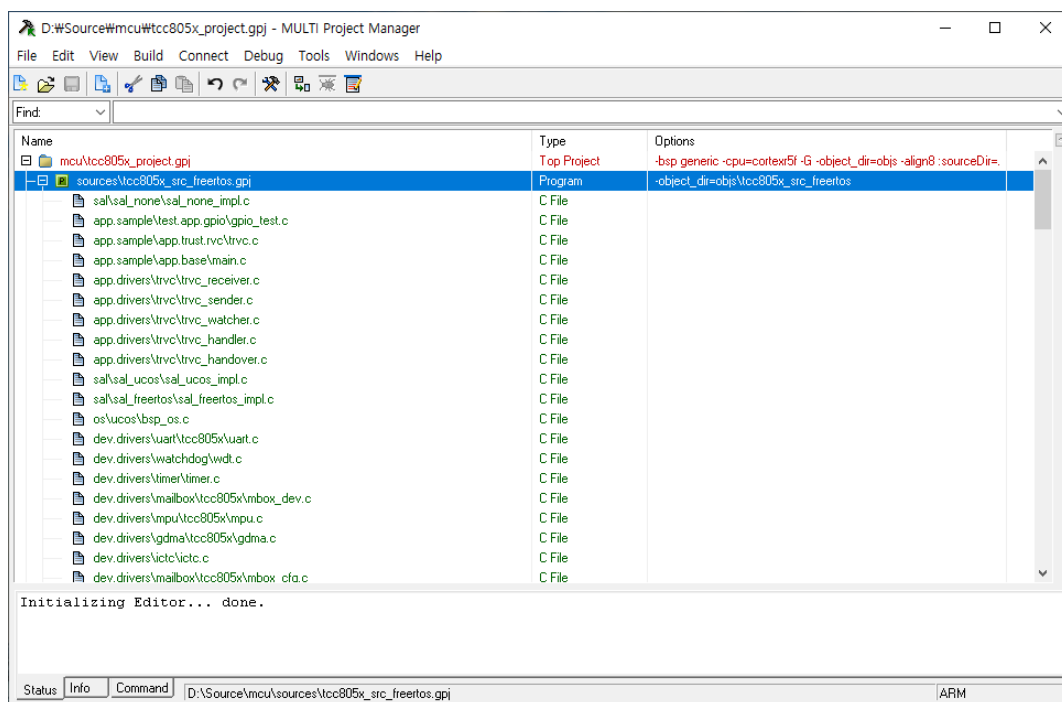


Figure 5.16 Complete Adding Source File

- 4) Add asm file: In **"Select Item to Add"** window → **"Add New or Existing"** → Add files in **"Other File"**.
 - Select the file type as **"Assembly Source"**.
 - Add a file.
 - The file extension distinguishes upper and lower case. Use lower case for .asm.

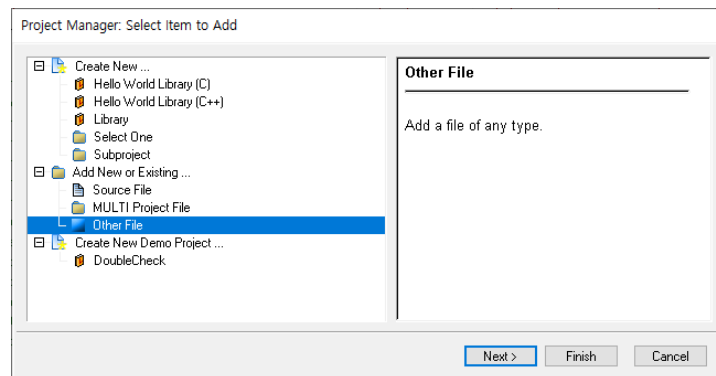


Figure 5.17 Add Assembly File (.asm)

- 5) How to add an include path
 - Designate **"include path"** to add a header file as `#include <header.h>`
 - Select a program → Right-click → Select **"Set Build Options"**.

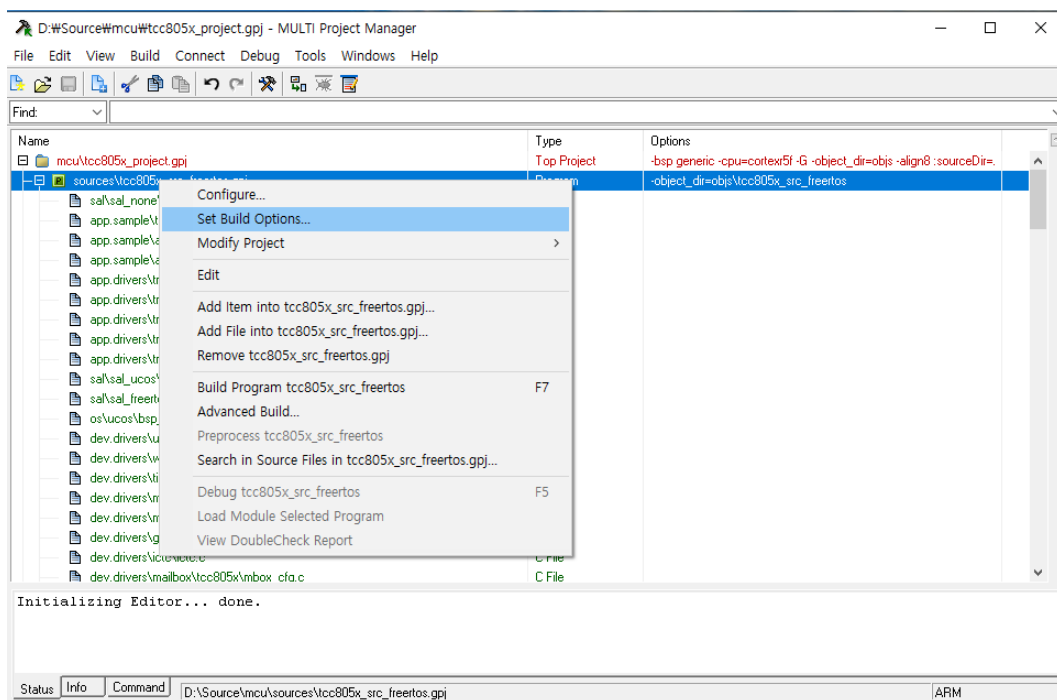


Figure 5.18 Set up Build Option

- Select **"All Options"** tab → Under **"Project"** → Select **"Include Directories"**.

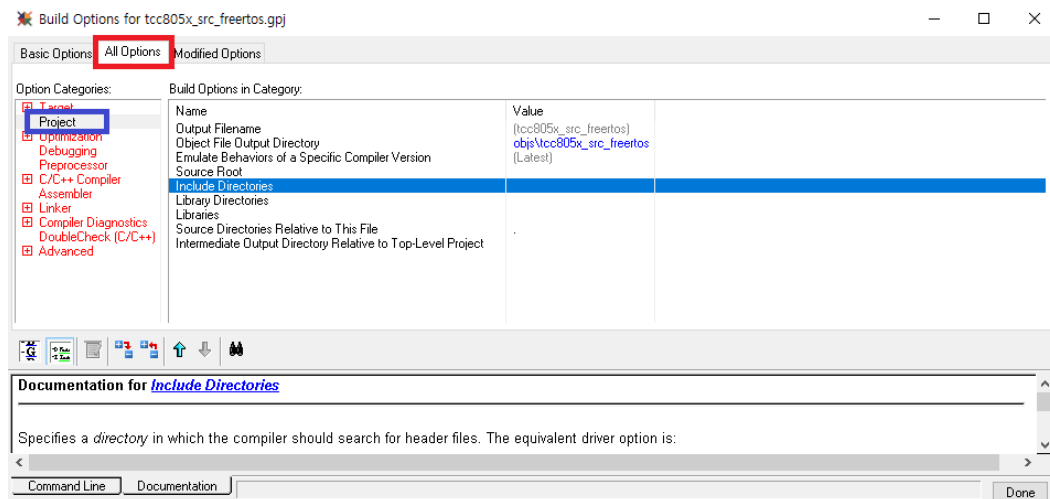


Figure 5.19 Configure Include Path

- Add new file path of a header file with relative path starting '\.' directive word. The include directories use the relative path of the header file from the location of the program file (tcc80x_src_freertos.gpj).
 - If you find the tcc805x_src_freertos.gpj file under the "\\build\tcc805x\ghc\src_program\" folder, see Figure 5.20 Add Include Path.

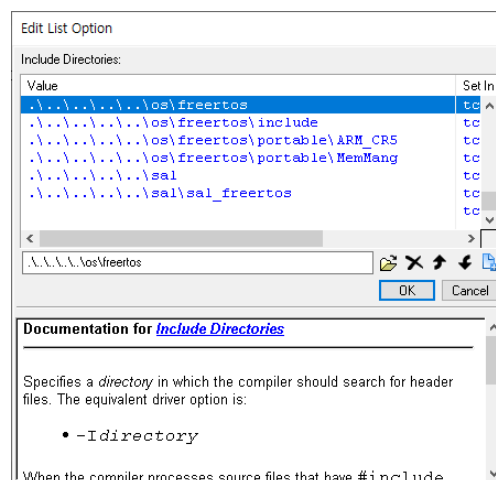


Figure 5.20 Add Include Path

- It is possible to add a path with **"Program".gpj** file through the text editor.

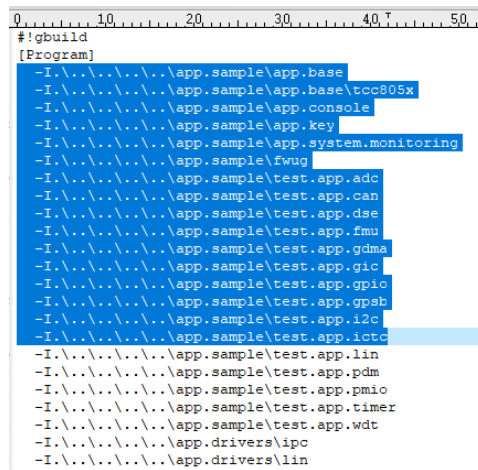


Figure 5.21 Add Include Path Manually

6) Memory Layout Setting

- Modify the basic .ld file. It is possible to select .ld file for each usage when compiling.
 - Select the program → Right-click → **"Configure"** → In **"Setting for Program"**, select the corresponding link option from **"Program Layout"**.

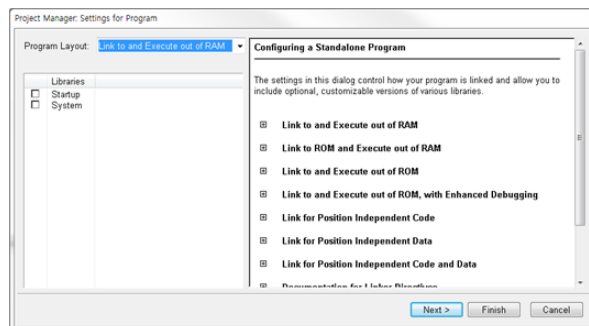


Figure 5.22 Set up Memory Layout

- It is possible to use .ld file written by you: Select "**Custom**" from "**Program Layout**" and the .ld file.
 - Using the S-NOR and Int.SRAM to copy "**Startup.s.data, .bss**" to the int.SRAM
 - standalone_romcopy tcc805x snor xx.ld

7) Output File Name Setting.

- When the 'Build' is normally completed, basically three files, "Program name".map (map file), "Program name" (no extension, elf file), and "Program name".mem (binary file), are generated.
- Changing "**Output Filename**": Under "**Program**" → Right-click → Select "**Set Build Options**" → All Options tab → Project → Click "**Output Filename**".

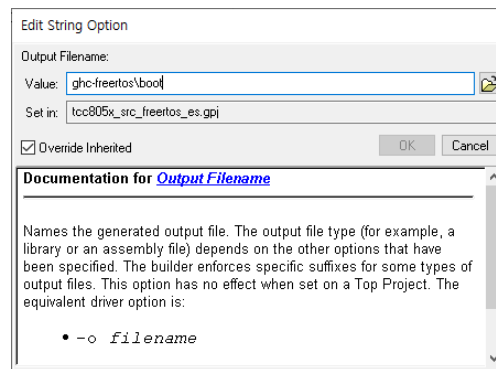


Figure 5.23 Output File Name

5.2.3 Build TCC805x MCU BSP

1. The MCU BSP Software provides the optimized project files of freeRTOS and uC/OS-III RTOS.
2. Open project file (tcc805x_ghc_freertos_xx.gpj) in "\\build\tcc805x\ghc" folder by using Green Hills MULTI IDE (In case of using uC/OS-III, use the tcc805x_ghc_ucos_xx.gpj file). [xx can be es (evaluation sample) or cs (customer sample)]
 - 1) Double click "**project file**".

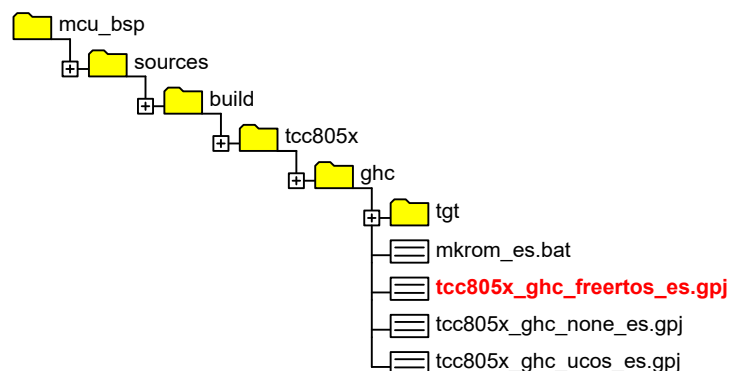


Figure 5.24 Open Project File

- 2) Modify EVB board definition in the source program file(.gpj) as below. The definition of the tcc805x EVB is set to "MCU_BSP_BOARD_VER_TCC8059_EVM_0_1" by default. If you use the TCC8050 EVB, you should set the definition to "MCU_BSP_BOARD_VER_TCC8050_EVM_0_1".

```
{optgroup=GhsCommonOptions} -o ghc-ucos\boot
-preprocess_assembly_files
-D__func__=__FUNCTION__
-DDMA_NC_SIZE=0x4000
-object_dir=ghc-ucos\obj
-cpu=cortexr5f
-DSNOR_STANDALONE
-DTCC805x
-DOS_UCOS
:postexecShellSafe='mkrom_cs.bat ucos'
-ON
-DMCU_BSP_BOARD_VER_TCC8050_EVM_0_1
-DPMU_DRIVER
-DDEBUG_ENABLE=1
-DCONSOLE_HISTORY_ENABLE
-D__MPU_ENABLED__
```

Figure 5.25 Modify EVB Definition

- 3) Select build menu: **"Build → Build program boot"** or use **[F7]** shortcut key.

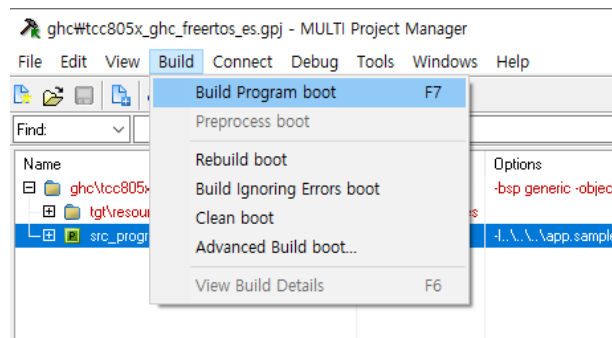


Figure 5.26 Build Project

- 4) Connect the executable binary image (boot.mem).

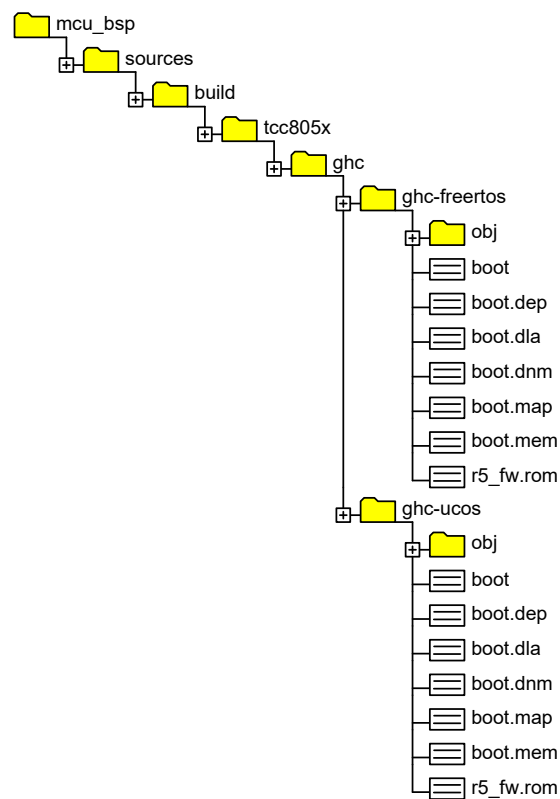


Figure 5.27 Build Completion

r5_fw.rom: Final MCU BSP rom file. It is created by adding a header to the binary file ("*boot.mem*"). It can be updated to the target EVB after being included in S-NOR image. Refer to the Chapter 5.3 Preparation Image for S-NOR Booting.

5.3 Preparation Image for S-NOR Booting

To update MCU BSP rom file, you should create an S-NOR image. It includes some headers and F/W images related to S-NOR boot and security. For details on how to create the S-NOR image, refer to the “*TCC805x Linux SDK-Getting Started* [References](#)”[2].

6 FWDN (FIRMWARE DOWNLOADER)

6.1 System Requirement

6.1.1 Software Requirement

Table 6.1 Software Requirement

Category	Description	Path	Note
VTC driver	<i>vtcdrv v5.0.0.14.zip</i> It is a driver for windows PC that is used to download firmware. VTC driver corresponding to the windows version must be installed. It is installed only once on your PC.	<i>tools/fwdn_v8</i>	You must use version 5.0.0.14 or higher.
FWDN	<i>fwdn.exe</i> Tool for downloading firmware.	<i>tools/fwdn_v8</i>	You must use version 8 or higher

6.1.2 Hardware Requirement

Table 6.2 Hardware Requirement

Category	Description	Note
EVB	Supported chipset are TCC805x	
Power Adapter	12V adapter is used to supply power to the board and LCD.	Recommend using the power adapter(12V/10A) provided by Telechips.
MicroAB port USB Cable	It is used to connect an EVB to PC for FWDN.	
USB C Type Cable	It is used to connect an EVB to PC for UART Log.	

6.2 Install VTC driver

Install USB driver (VTC driver V5.0.0.14) for FWDN in the windows PC. The driver can be installed only once on PC. The VTC driver V5.0.0.14 is located in "tools/fwdn_v8".

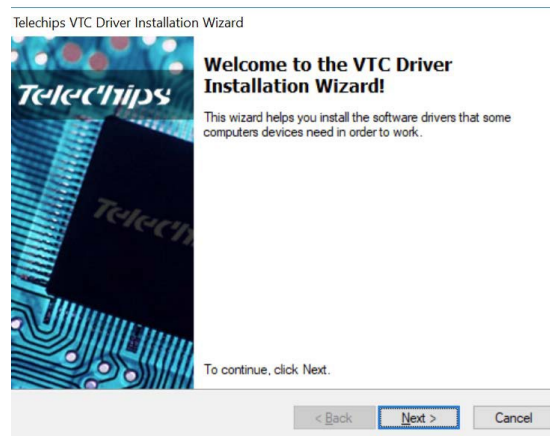


Figure 6.1 VTC Driver Install Window

6.3 FWDN Sequence

1. Change boot mode switch to USB boot mode and TCC805x EVB power off → on.
2. Move "tools/fwdn_v8" directory.
3. Execute FWDN and download S-NOR image. You can select executing FWDN command directly or using batch file.
 - 1) FWDN command
 - i. fwdn.exe --fwdn tcc805x.fwdn.json
 - ii. fwdn.exe --write tcc805x_boot.json.
 - iii. fwdn.exe --write tcc805x_snor_boot.rom --area die1 --storage snor
 - 2) Sample batch file
 - i. sample_download_image2snor.bat: update S-NOR image
 - ii. sample_low_format: erase S-NOR and eMMC memory
4. Change boot mode switch to S-NOR boot mode and TCC805x EVB power off → on.
5. Check whether MCU BSP is running via UART log. If you want to see log below, change "MCU_BSP_BUILD_FLAGS_DEBUG" flag to 1 from 0 in Makefile.

```
Initialize System done
Welcome to Telechips MICOM
PowerDown reason is watchdog

MICOM BSP Version: V1.0.0
(D)[SAL ][FR_OSStart:543] ~ Done to initialize Free RT Operating System ~

-----TCC_EVB_BD_VERSION == TCC8059_BD_VER_0_1-----
```

Note: The files for FWDN are included in boot-firmware. For detailed information about FWDN and boot-firmware, refer to the "TCC805x Linux SDK-Getting Started"[2].

7 HOW TO DEBUG BY USING JTAG DEBUGGER

You can debug software by using JTAG debugger device such as Trace32, CodeViser and so on. The script for Trace32 is supported but it is not official. For more detailed information about Trace32 script, contact Trace32 supplier. TCC805x MCU BSP was tested under the following conditions.

- Hardware: TCC805x EVB SV0.1 (CPU Board) and CV0.11 (Main Board)
- Software: TCC805x MCU BSP v1.00
- Compiler: GCC
- JTAG debugger: Trace32

The sequence below shows debug method using Trace32 scripts.

1. Connect JTAG probe of JTAG debugger device and JTAG connector of TCC805x EVB.
2. Change boot switch to USB mode.
3. Download "tcc805x_snor_boot.rom" via FWDN.
4. Change boot switch to S-NOR and eMMC boot mode.
5. Reboot TCC805x EVB.
6. Run Trace32 PowerView (Trace32 PC program).
7. Open script (Trace32 PowerView → File → Edit Script → select "*mcu_bsp/scripts/debug/T32/tcc805x_nor_attach.cmm*")
8. Edit image path and save.
9. Click **"Do"** button.

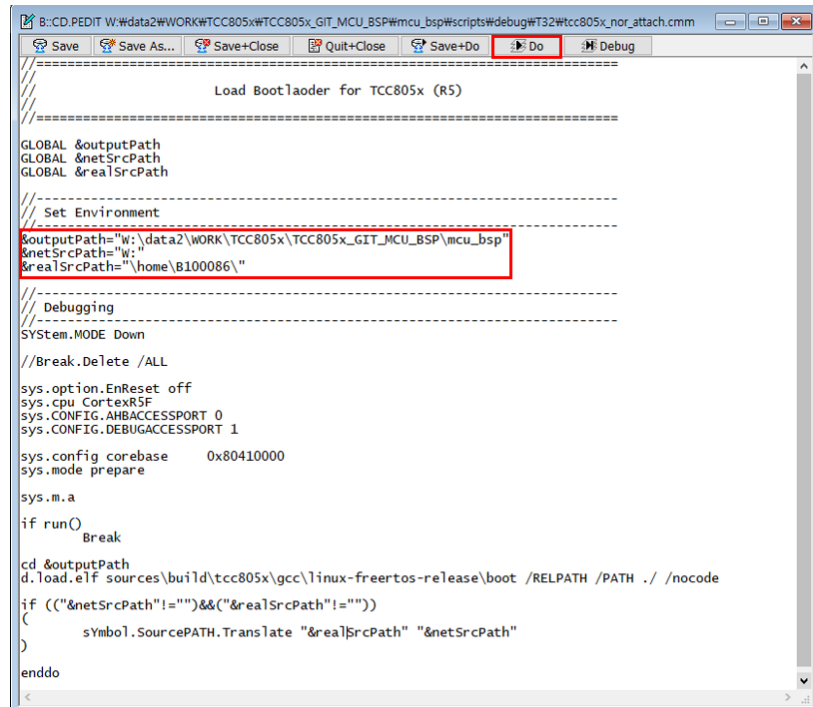


Figure 7.1 Trace32 Script for Debug

You can debug software if above sequence is normal. For more details about Trace32, see the manual of Trace32.

8 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.01: 2021-04-09

- Updated
 - Chapter 5.2.3: Description to modify EVB definition on the Green Hills IDE
- Added
 - Chapter 5.2.3: Figure 5.25

Rev. 1.00: 2020-11-12

- First version release

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